

Execution Style Clustering On Polymarket Does Not Recover Scalable Skill

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Abstract:

Standard financial analysis usually looks at trader performance through 2D charts like PnL graphs or book depth charts. In this project, I took a different approach by treating Polymarket trade streams as a 3D point cloud. I mapped 600,421 Polymarket traders into a 3D execution space (price book impact, time-of-day dispersion, and log trade size). Drawing on my background with 3D LiDAR processing, I applied HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) without using PnL. The fit yields 1,684 dense groups/clusters and 62.3% noise. In this dataset, “sharp” (a PPV bin, not an independent skill tag) is only 17.6% of dense points vs 15.7% of noise. The densest “sharp” group is ~1500 traders with ~1 trade and ~\$1 volume. Execution similarity tracks how people trade (often small, specialized, same clip size), not whether they have a scalable edge. A 3D density model is useful for finding behavioral similarities, but not, by itself, an alpha screen.

How I Began:

I began my research by opening the Polymarket data in a Parquet reader to analyze the columns present so I could brainstorm a hypothesis. Through some quick Google searches, I learned that Polymarket data is usually visualized in 2D: order book depth charts, price-over-time line graphs, etc. I wanted to build on my previous experiences and think of a topic that could play to my strengths but also have a level of novelty and innovation involved. This led me to the following idea:

1. 3D Volumetric Clustering of Market Microstructure: Applying spatial point cloud mechanics (LiDAR tiling and density) to the prediction market order book.
 - a. Having worked with LiDAR data previously and handling millions of points distributed across 3D coordinates, I noticed an order book and trade stream is essentially a 3D point cloud: (X = book levels consumed, Y = volume-weighted dispersion of time-of-day, and $Z = \log_{10}(\text{mean_tx_value} + 1)$). **Could we examine these point clouds to determine and test whether repeating execution geometry is associated with higher PPV in the dataset sample?**

Testing this hypothesis could allow me to see if sharp traders form repeating geometric patterns and if “bad” or “awful” traders are more scattered and mathematically random. Essentially, I tested whether repeating execution geometry lines up with higher in-sample PPV.

Step 1: Data Preprocessing/Cleaning

The trader-level records were read from a parquet file (N = 604,578 unique traders) using pandas with the pyarrow engine. I required the following fields:

1. **identity** fields *trader* and *trader_label*
2. the **performance** fields *trader_pnl*, *trader_volume*, and *trader_ppv*
3. the clustering **coordinates** (X, Y, Z) *price_levels_consumed_vw*, *std_time_vw*, and *mean_tx_value*
4. the set of **category-share** columns that begin with *topic_* (17 columns total in this dataset)

If any *trader_label* values were missing, they were recorded with “Unlabeled”. Non-finite values in the coordinate and performance fields were treated as missing. Rows lacking any of those six numerical fields (in coordinate or performance fields) were dropped listwise (n = 4,157, 0.69%), almost entirely because *std_time_vw* was missing or infinite. Missing data was not replaced with averages, medians, or placeholder values as a missing 3D coordinate would invent a location in the cluster space and break performance overlays. They were simply removed. To stabilize heavy-tailed distributions (e.g. whales, who have enormous total trading volume and mean transaction value) for Euclidian clustering, I retained the raw variables and added $\log_{10}(\text{mean_tx_value} + 1)$ and $\log_{10}(\text{trader_volume} + 1)$.

This compresses large gaps so different size traders can be grouped fairly. Adding 1 also handles zero-volume or zero-transaction edge cases and avoids undefined limitless $\log(0)$.

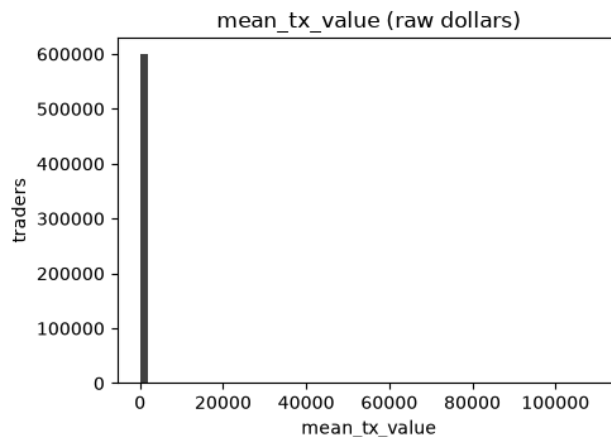


Figure 1: Raw *mean_tx_value*. A few very scattered whales skew the data heavily.

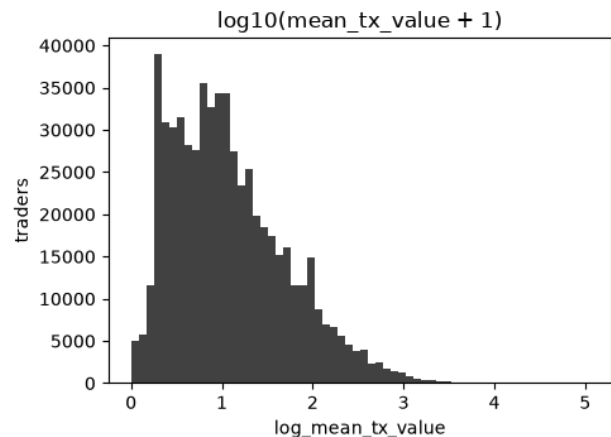


Figure 2: Log transformed *mean_tx_value*.

The resulting analysis sample is N = 600,421 traders and 43 columns.

Step 2: Feature Engineering

To capture behavioral differences across traders, I engineered two sets of features prior to clustering: a topic concentration index to quantify specialization, and a standardized 3D feature matrix to capture trader execution styles.

2.1 Topic Specialization Index (HHI)

I used the Herfindahl–Hirschman Index (HHI) to measure how focused or diversified a trader is across topics by squaring each topic column share and summing them together. The

Herfindahl–Hirschman Index is calculated for each trader as: $HHI_i = \sum_{c=1}^{17} s_{i,c}^2$

Because each topic share $s_{i,c}$ is a fraction between 0 and 1, squaring the fractions gives more weight to higher numbers. This allows us to separate topic specialists (traders who focus on a single topic/domain) with generalists (traders who trade across all different topics), regardless of how much total volume they trade.

2.2 Execution Coordinates + Log Transformation

Like I mentioned in Section 1, to group traders by *how* they trade rather than *what* they trade, I selected three execution metrics:

1. *price_levels_consumed_vw*
2. *std_time_vw* (how spread out trades are over time, volume-weighted)
3. *mean_tx_value* (average trade size)

Only the average trade size is logged. Clustering uses raw *price_levels_consumed_vw*, raw *std_time_vw*, and $\log_{10}(\text{mean_tx_value} + 1)$. *log_volume* is computed and unused.

2.3 Rescaling Data (Z-Score Standardization)

Density-based clustering algorithms like DBSCAN and HDBSCAN (the latter in this scenario) group data by measuring straight-line (Euclidian) distances between points in space:

$$d(p, q) = \sqrt{\sum_{k=1}^3 (p_k - q_k)^2}$$

Because the features operate on fundamentally incompatible scales (e.g. *std_time_vw* is measured in milliseconds (inferred), while *log_mean_tx_value* ranges between 0 and ~5.04), if fields were unscaled then the distance calculations would be almost entirely driven by time.

To give all three metrics equal weight, I standardized each feature using StandardScaler (Z-score normalization)

$$x' = \frac{x - \mu}{\sigma}$$

This centers each metric at 0 with a standard deviation of 1, ensuring that time, order book depth, and trade size contribute equally when grouping similar traders.

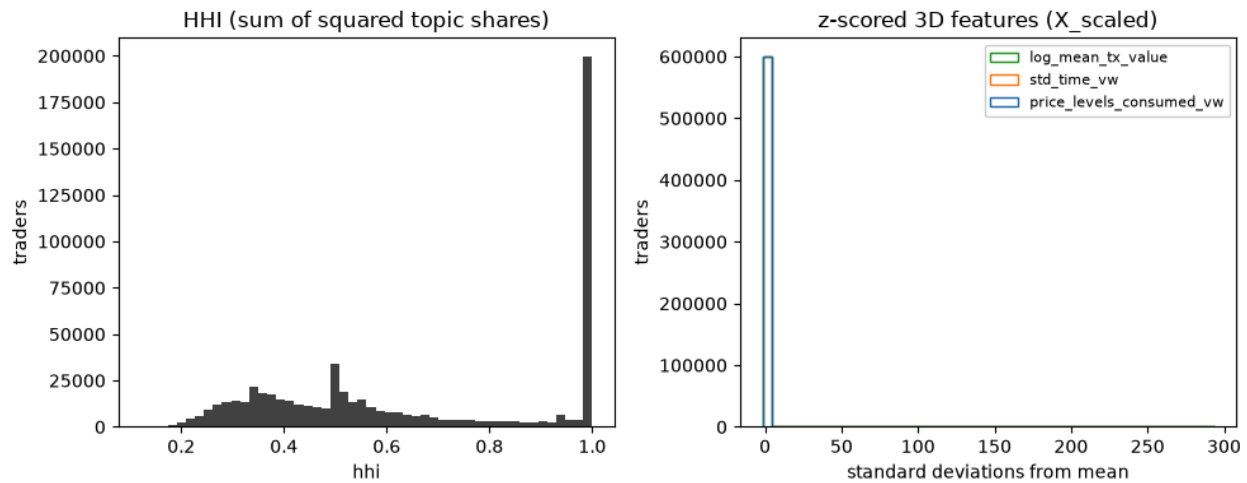


Figure 3: The HHI of 1 represents pure specialists who only trade in one topic. Everything to the left is a generalist.

Figure 4: Distribution of the three axes after standardization. As shown, they are all on the same scale (Table 3).

Step 3: HDBSCAN Spatial Clustering + Anomaly Isolation

To discover distinct trader execution styles without forcing every trader into an arbitrary category, I applied HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) to the standardized 3D coordinate system.

3.1 Why HDBSCAN over k-means?

Unlike distance algorithms like k-means, which forces every data point into one of k predefined clusters, HDBSCAN identifies regions of high point density while classifying sparse or unstructured data as noise (`spatial_cluster = -1`). In financial markets, where many traders execute opportunistically or randomly, HDBSCAN can allow us to isolate structured execution patterns while ignoring background noise.

3.2 Parameter Configuration

The HDBSCAN algorithm relies on two primary tuning parameters:

- `min_cluster_size = 50`: I set this to 50 traders (about 0.008% of the dataset) because I needed a solid floor for what counts as an actual cluster. If I set it any lower, the model starts picking up random groups of 3 or 4 traders and treating them like distinct trading strategies, which introduces way too much noise.
- `min_samples = 15`: This acts as my local density filter. I required a trader to have at least 15 close neighbors in 3D space to be marked as a core cluster member. I picked 15 as a middle ground, as it's strict enough to filter out random retail trades, but relaxed enough that it doesn't drag isolated outliers into clusters.

3.3 Output Features + Dataset Integration

Fitting HDBSCAN on the cleaned sample ($N = 600,421$) generated two primary output metrics attached directly to each trader's row:

1. `spatial_cluster` (cluster identifier)
 - a. ≥ 0 : assigned to one of the 1,684 dense cluster regions, representing traders who follow repeating execution patterns in 3D space
 - b. `-1`: assigned to unstructured spatial noise (373,773 traders, or 62.3%).
 - i. **IMPORTANT**: Being labeled as noise does not mean a trader is unskillful; it simply indicates that their execution style does not belong to a high-density cluster.
 - ii. 1,684 small clusters shows the market might contain many specialized micro-strategies rather than just 2 or 3 broad strategies/archetypes.
2. `outlier_score` (GLOSH Outlier Score):
 - a. A bounded score $[0,1]$ measuring local sparsity using Global-Local Outlier Scores from Hierarchies (GLOSH). Scores near 0 represent central, representative cluster members while scores near 1 denote sparse points/edge points.

3.4 Decoupling Style from Performance

Performance metrics (PnL, profit per dollar traded, and trader classifications) were intentionally excluded from the clustering process as clustering was performed exclusively on execution coordinates (*price_levels_consumed_vw*, *std_time_vw*, and *log_mean_tx_value*).

By aggregating performance metrics after cluster labels were assigned, we evaluated whether specific structural trading styles systematically outperformed the larger background noise pool.

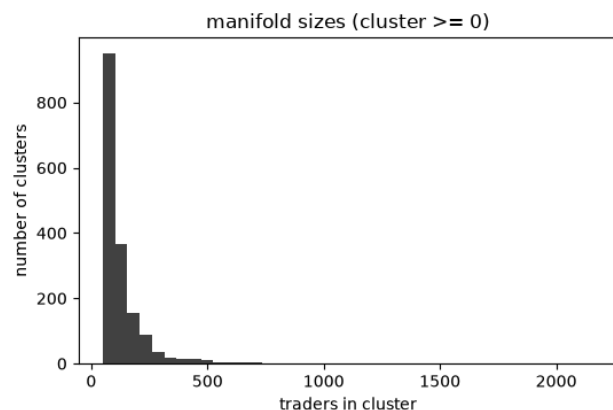


Figure 5: Sizes of the 1,684 clusters detected by HDBSCAN. Most clusters contain between 50-200 traders, with a long right tail extending up to ~2000 traders.

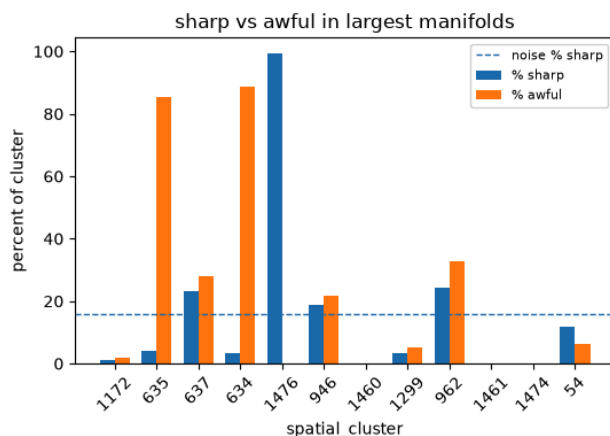


Figure 6: Proportion of sharp vs. awful traders across 12 selected large clusters. The dashed baseline represents the sharp trader percentage in the background noise pool (~16%). Noticeable performance divergence exists across clusters, e.g. cluster 1476 is ~99% sharp whereas clusters 634 and 635 are >85% awful.

Step 4: 3D Visualization Pipeline

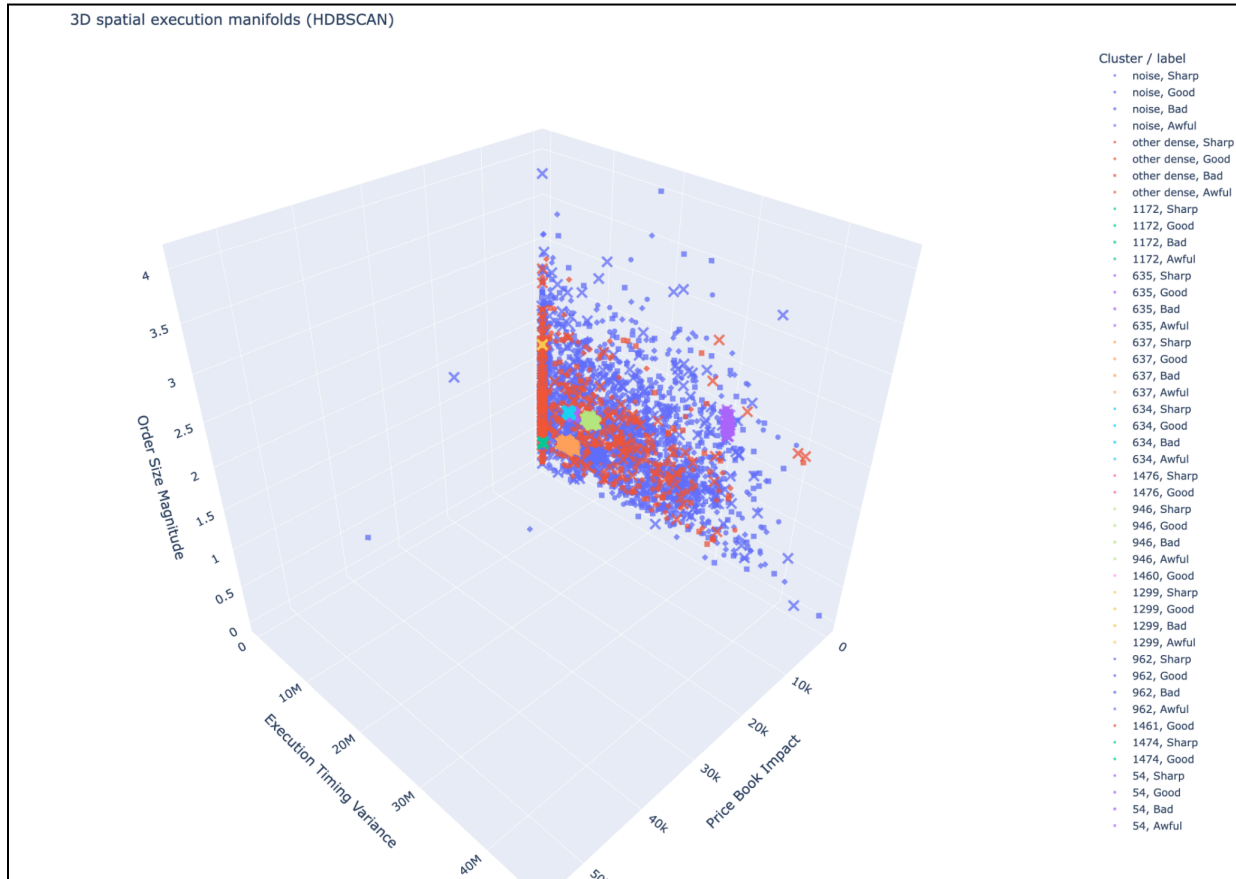


Figure 7: Downscaled 3D map of Polymarket trader execution style. Axes are $\text{price_levels_consumed_vw}$, std_time_vw , and $\log_{10}(\text{mean_tx_value} + 1)$. Color is the HDBSCAN spatial cluster treated as a categorical label (12 largest clusters shown in distinguishable colors, remaining dense points are grouped as other dense, and unstructured points are noise). The marker symbol is the provided trader label (sharp, good, bad, awful). Hover fields report trader_pnl , trader_ppv , and hhi . For readability, the figure displays a partitioned subsample of the total data: largest clusters and a slice of other dense points, and a cap on noise. Cluster membership itself was estimated on the full z-scored matrix. After z-score, ~97% of points sit on one book impact plane (Table 3) so if noise is hidden the cloud is visually a flat pancake along X.

Full Dataset 3D Viewer: <https://lilys7.github.io/polymarket-analysis/web/>

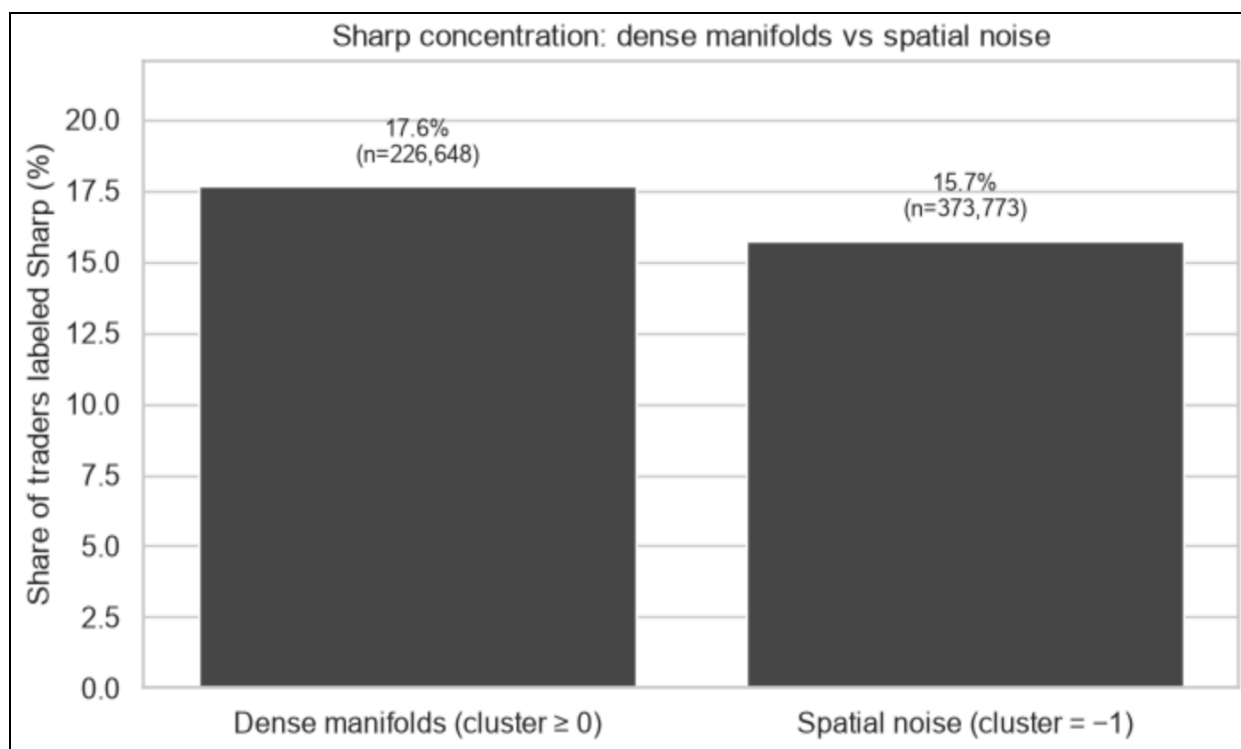


Figure 8: Share of traders labeled “sharp” in dense HDBSCAN clusters ($\text{spatial_cluster} \geq 0$) vs. the unstructured spatial noise pool ($\text{spatial_cluster} = -1$). In the full clustered sample ($N = 600,421$), “sharp” accounts for 17.6% of dense-cluster traders ($n = 226,648$) and 15.7% of noise ($n = 373,773$). The gap is pretty small: labeled high-PPV traders are only mildly over-represented in the detected clusters.

In short: “sharp” is just a math label for high PPV in this dataset, not a verified measure of long-term trading intelligence. Therefore, the slight difference in percentages shows a pattern in the data, but it does not prove that structured execution styles make traders inherently more skilled.

Step 5: Results + Discussion

5.1 Hypothesis Evaluation:

I expected dense 3D clusters to be “sharp” and noise to be “awful” or “bad”. In other words, structured execution = skill.

5.2 What Actually Happened:

“Sharp” is 17.6% of dense clusters vs 15.7% of noise. The gap is ~2 pp. 66.4% of “awful” + “bad” traders sit in noise, but 62.3% of ALL traders sit in noise, so that is only a +4.2 pp over the base rate. “Sharps” are 59.5% in noise, the same as “awfuls” (59.5%). Density does not separate skill from noise.

Table 1: Cluster Breakdown

Table 1. Cluster autopsy: dense Sharp / Awful manifolds vs spatial noise

cluster	n	median trades	median volume	median PPV	% Sharp	% Awful
1476	1,543	1	\$1.04	+0.0400	99.4%	0.1%
634	1,705	2	\$43.71	-0.3820	3.2%	88.7%
635	1,850	2	\$44.77	-0.3675	4.2%	85.3%
noise (-1)	373,773	8	\$228.00	-0.0009	15.7%	19.8%

Table 1: Cluster 1476 (n=1,543) is 99.4% “sharp” because it is 1,543 traders with a median of 1 trade and a median volume of \$1.04. Std_time_yw = 0 here is the std of one timestamp, not a TWAP bot. Clusters 634 and 635 (n=1,705 / 1,850) are 88.7% and 85.3% “awful”, median 2 trades, median volume ~\$44, median PPV ≈ −0.38 / −0.37. HDBSCAN found similar trade size clones, not scalable alpha.

Table 2: Base Rates

Table 2. Share of each PPV label sitting in spatial noise vs the 62.3% base rate

label	n	n in noise	% of label in noise	vs 62.3% base rate
Awful	124,718	74,173	59.5%	-2.8 pp
Bad	197,007	139,572	70.8%	+8.6 pp
Good	179,879	101,198	56.3%	-6.0 pp
Sharp	98,817	58,830	59.5%	-2.7 pp
Awful + Bad	321,725	213,745	66.4%	+4.2 pp
all traders	600,421	373,773	62.3%	+0.0 pp

Table 2: Only “bad” is clearly overrepresented in noise (+8.6 pp). It cannot be determined that losers are scattered in a way “sharps” aren’t.

Table 3: Book Impact

Table 3. Book-impact axis is degenerate: one clustering coordinate barely moves

statistic	value
share with book impact exactly 0	68.9%
median book impact	0
z-score of the zero pile	-0.034
share within 0.05 z of the zero pile	97.1%
99th percentile of z(book)	0.33
max z(book)	293
median z(std_time_vw)	-0.14
median z(log_mean_tx_value)	-0.16

68.9% of traders have book impact exactly 0. 97.1% sit within 0.05 z of that pile. Clustering is mostly std_time_vw and log clip size.

5.3 Limitations:

The following are the main limitations of this study:

1. No trade-level time in the dataset, so *future* PPV cannot be tested/predicted.
2. “Sharp” is a PPV bin, not a grade for skill of traders.
3. The file has no bot or maker/taker flag. Cluster 1476 is 99% “sharp”, but it is 1,543 traders with median 1 trade and ~\$1 volume, so it cannot be read as traders using algorithmic execution (e.g. TWAP/VWAP)
4. One-trade traders force *std_time_vw* to 0 (std of a single timestamp). HDBSCAN treats that as a dense “style” even though it’s just one trade; looks like super tight timing.

5.4 Takeaways:

Repeating how someone trades (size, clock spread, book impact) is **NOT** the same as being a good trader. Dense clusters observed in the 3D model are simply style clones. The notable 99% “sharp” cluster is ~\$1 one-trade tickets. The skill-looking labels barely differ in dense vs noise (as shown in Figure 8). Noise is not just a bunch of random losers as “sharps” are prevalent there at about the same rate as “awfuls.”

In conclusion, the 3D model is good at finding traders that execute the same behaviorally. It does not let us detect edge.